

SEHAJ SINGH

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EDUCATION

University of Maryland, College Park

College Park, MD

BS in Computer Engineering, Minor in Entrepreneurship

Expected May 2028

- Coursework: Computer Systems, Data Structures and Object Oriented Programming, Discrete Structures, Digital Logic Design, Signals and Image Processing, Linear Algebra, Differential Equations

TECHNICAL SKILLS

Languages: C++17, Python, C, TypeScript/JavaScript, Java, SQL, MATLAB

Graphics & Perception: OpenGL/GLSL, Gaussian Splatting, SAM 3, MuJoCo, PyTorch, OpenCV, NumPy, SciPy

Systems & Web: React, Next.js, Node.js, FastAPI, WebSockets, WebRTC, MongoDB, OpenSearch, Git, Linux

EXPERIENCE

Paphos Oneironautics

June 2026 – Sept. 2026

Software Engineer Intern

Remote

Robotics startup building a cable-driven farming robot. I turned a 3D scan of a real orchard into individual trees and apples the robot can find, count, and drive to.

- Built a SAM 3 open-vocabulary 3D instance segmentation pipeline, projecting Gaussians into 85 posed views and resolving cross-view identity by mask overlap, reaching F1 84.2 vs 69.3 baseline
- Deduplicated multi-view fruit detections by cross-view SAM-mask consistency, reaching precision 0.91 (F1 0.81) against the dataset's 105 hand-annotated apples
- Built any-angle (Theta*) path planning over the segmented voxel scene, constraining search to the robot's reachable cable plane: mean waypoints 59 to 2.1, path cost down 32%, worst-case node expansions 154,000 to 458
- Shipped a natural language command bar into a C++ flecs engine: an offline parser with an Anthropic API fallback on a worker thread

AAC Inc.

June 2026 – Aug. 2026

Software Developer Intern

Tysons, VA

- Root-caused two clamping bugs in a Python telemetry engine, cutting erroneous records from 566,708 to 0 across a 5M-record dataset
- Proved a third bug with a controlled A/B test, forcing two servers into different states and using OpenSearch std dev (0.031 vs 4.19) as evidence
- Built and operated a multi-server data pipeline on AWS EC2 across 3 Linux hosts, provisioning environments, writing bulk-upload automation, and scheduling nightly cron backfills

University of Maryland, Perception and Robotics Group

Feb. 2026 – Present

Undergraduate Research Assistant

College Park, MD

- Train and export Gaussian Splatting and NeRF models on a SLURM-scheduled GPU cluster, benchmarking reconstruction quality across competing methods
- Got an Instruct-GS2GS pipeline running end to end by resolving version conflicts across the HuggingFace and CUDA stack on cluster hardware
- Implement least squares estimation and sensor calibration algorithms for high-dimensional data in real-time robotic systems

PROJECTS

Shadow Box | TypeScript, React, Node.js, WebSockets, WebRTC, MediaPipe

shadowbox.gg

- Shipped and operate a live real-time 1v1 multiplayer browser game with peer-to-peer WebRTC video
- Wrote the netcode: a WebSocket server holds authoritative match state and resolves simultaneous move commits
- Built in-browser MediaPipe pose detection scoring player gestures at framerate; own the stack through deployment

Internia | Next.js, TypeScript, FastAPI, MongoDB

- Co-built an ML research paper recommender on the Semantic Scholar API, filtering 1M+ papers across 24 domains
- Built the Next.js and TypeScript frontend with NextAuth on a FastAPI and MongoDB backend

LEADERSHIP

Project LIFT

May 2026 – Present

Executive Board, Client & Project Operations

College Park, MD

- Co-lead client operations for a 60+ student org that has delivered \$60,000+ in software value to local small businesses
- Run discovery calls and author scope, fit, and risk decision docs determining which clients the org takes on, targeting 8 for the year